

Cluster Around and Find Out: A Conversational Clustering System

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June 4, 2026

Abstract

Evaluating text clustering solutions is inherently challenging due to the subjective nature of human preferences and the absence of a universal ground-truth assignment. This paper introduces *Cluster Around and Find Out*, an interactive, conversational text clustering system that leverages large language models (LLMs) to elicit user preferences through multi-turn dialogue. Rather than evaluating the underlying clustering algorithm in isolation, this study focuses on the quality, efficiency, and conversational dynamics of preference elicitation. We design a controlled experimental framework ($N = 108$ runs) simulating distinct user personas and goals to evaluate three interaction paradigms: yes/no, multiple-choice, and open-ended questions. Our primary outcome tracks the conversational turns required to achieve convergence, complemented by secondary measures of token efficiency and cluster faithfulness evaluated via an LLM-as-judge. We hypothesize that multiple-choice queries offer the optimal tradeoff between human cognitive load and information gain per turn. Finally, we compare our interactive approach against an uninstructed baseline to isolate the impact of preference-driven embedding guidance.

1 Introduction

Unsupervised text clustering serves as a cornerstone for exploring unlabelled textual data. How-

ever, traditional algorithmic approaches operate under a rigid paradigm, assuming that a singular, objectively correct group assignment exists [1, 2]. In practical applications, text categorization is highly subjective; two users analyzing the same dataset may demand completely divergent cluster structures based on their immediate task constraints, domain expertise, or semantic perspectives [3]. Consequently, static clustering solutions often fail to align with idiosyncratic human intents.

To address this gap, recent paradigms have shifted toward integrating human-in-the-loop guidance within clustering workflows. Concurrently, Large Language Models (LLMs) have emerged as powerful conduits for natural language preference elicitation, uncovering latent user nuances far more dynamically than passive feedback mechanics [3, 4, 5]. By positioning an LLM as an interviewing agent, a system can iteratively converse with a user, capture their architectural preferences for the data, and adapt downstream clustering behaviors accordingly.

This project introduces *Cluster Around and Find Out*, a conversational text clustering system designed to bridge user intent and document representation. While clustering remains central to our pipeline, our empirical evaluation targets the conversational interface itself rather than raw cluster geometry against artificial baselines. Specifically, we investigate two core research questions:

- **Q1:** Which questioning paradigm (yes/no,

multiple-choice, or open-ended) yields the most efficient path to convergence while minimizing human cognitive load?

- **Q2:** Does incorporating task-specific instructions derived from conversational preferences materially improve the subjective faithfulness of the resulting clusters compared to an uninstructed baseline?

We formalize our investigation through a rigorous study design utilizing a simulated user with 3 distinct user personas and 6 specialized clustering goals across a subset of the StackOverflow dataset. We explicitly test the hypothesis that multiple-choice questions strike the premier equilibrium—optimizing the volume of preference data harvested per turn without placing excessive linguistic or cognitive demands on the user.

The remainder of this report is organized as follows: Section 2 contextualizes our approach within the state-of-the-art literature; Section 3 and Section 4 outline our experimental variables and system architecture; Section 5 and Section 6 present our bootstrap-tested findings; and Section 7 addresses key confounding factors and future avenues.

2 Related Work

The intersection of large language models (LLMs) and text clustering has catalyzed a shift from purely statistical grouping to semantically driven, human-aligned categorization. This section contextualizes our system within recent methodological precedents spanning conversational preference elicitation, LLM-guided clustering mechanics, and instruction-tuned text embeddings.

2.1 Conversational Preference Elicitation

Traditional preference learning often relies on passive feedback or static pairwise interactions. Recent studies demonstrate that multi-turn natural language dialogues significantly increase informative value, reduce user friction, and expose

nuanced criteria that passive settings miss [3]. To optimize these dialogues, existing frameworks frequently depend on complex architectural additions or expensive post-training phases. For instance, TO-GATE relies on trajectory optimization to balance clarification resolvers with task-aligned responses [4], while other approaches leverage diffusion-inspired funneling strategies to fine-tune LLMs for incremental profile generation [5].

In contrast to these computationally heavy fine-tuning and optimization methods, our work introduces a streamlined, production-ready prompting paradigm. We exploit the strong instruction-following capabilities of Qwen3.6-35B-A3B [6] by engineering a robust system prompt that transforms the model into an adaptive interview agent. By feeding initial cluster insights directly into this system prompt, the agent is dynamically empowered to guide the conversation and extract actionable user preferences without requiring any underlying model fine-tuning. This demonstrates that sophisticated, context-aware preference elicitation can be achieved purely through strategic prompt engineering and contextual data conditioning.

2.2 LLM-Assisted Clustering Algorithms

Rather than treating clustering as a separate downstream task, modern frameworks inject LLM intelligence directly into the clustering loop. Early methods demonstrated that foundational models could act as zero- or few-shot categorizers by leveraging in-context learning to assign text fragments to candidate clusters based solely on illustrative prompt examples [7]. Building upon this, ClusterLLM utilizes hard triplet questions (e.g., assessing relative document similarities) and pairwise granularity queries to guide hierarchical text partitioning [1]. However, traditional pairwise constraints scale poorly on extensive datasets. To alleviate this, optimized algorithms generate multi-element constraint sets (Must-Link and Cannot-Link blocks) via LLMs, achieving massive reductions in total API queries [8].

Other paradigms replace numeric geometric

operations entirely. *Dial-In LLM* incorporates an LLM judge to label cluster coherence and iteratively isolate stable partitions while merging redundant configurations based on text intent labels [2]. Furthermore, approaches like *k-LLMmeans* replace traditional numeric vector centroids with concise, LLM-generated textual summaries, enriching semantic interpretability and coherence [9]. In extreme cases, frameworks such as LLM-MemCluster eschew vector spaces altogether, manipulating a dynamic memory set of cluster labels with linear runtime complexity [10].

2.3 Instruction-Guided Embeddings

While memory-only and summary-centroid systems offer distinct advantages, vector representations remain crucial for scalable text matching. Crucially, a fixed embedding space cannot capture subjective, task-specific groupings. Recent breakthroughs demonstrate that decoder-only LLMs can be transformed into generalist embedding models by replacing causal masking with bidirectional attention and integrating pooled latent layers [11].

Models such as *e5-mistral-7b-instruct* prove that embedding vectors can adapt dynamically to explicit user instructions passed directly into the encoder prompt [12]. In this project, we utilize **Qwen3-Embedding (8B)** [13], an advanced instruction-tuned encoder that leverages this exact paradigm. By encoding the rich task instructions derived from our Qwen3.6-35B-A3B interviewing agent, the Qwen3-Embedding model projects document vectors into a space that reflects the user’s subjective criteria, modifying semantic distances dynamically without requiring downstream fine-tuning.

2.4 Methodological Gap: Subjective Alignment

A critical limitation within current literature is the pervasive assumption that an objective, underlying “ground-truth” assignment exists for every text corpus, typically validated using external metrics like Normalized Mutual Information

(NMI) or Rand Index (ARI) [1, 7, 10, 8].

Our work departs from this paradigm by addressing real-world deployments where text clustering is highly subjective and rigid mathematical benchmarks often misalign with a user’s actual downstream goal. Because no single, universally correct cluster assignment exists for such tasks, there is a distinct scarcity of research surrounding purely user-aligned evaluation. Consequently, this study shifts the evaluative focus away from geometric benchmark matching. Instead, we formalize internal conversational metrics—such as turn efficiency, conversation length, and LLM-as-judge multi-dimensional faithfulness—to assess how effectively a system adapts its representations to a user’s idiosyncratic intent.

3 Study Design

To systematically investigate the dynamics of interactive, conversational preference elicitation, we formalize our experiment using a multi-factor simulation environment. This design maps user interaction types against varied user objectives to evaluate the structural bounds of our system.

3.1 Factor Structure

Our experimental design consists of independent variables, observed dependent variables, and tightly controlled constants.

- Independent Variables (Manipulated):** The primary manipulated factor is the *question type* prompt-engineered into the Qwen3.6-35B-A3B interviewing agent. We test three distinct interactive formats: *yes/no* questions, *multiple-choice* questions, and *open-ended* questions. Additionally, the internal directive of the simulated user changes systematically on each run across 3 different personas and 6 unique goals to test the adaptive boundaries of the agent.
- Dependent Variables (Observed):** For each individual run, we measure conversational efficiency via the total number of

turns to convergence and the total volume of tokens exchanged. Question clarity is tracked via the *user response token length*. Subjective accuracy is captured via a multidimensional *faithfulness score* computed using an LLM-as-judge.

- **Constants:** To guarantee that observed changes are driven solely by the independent variables, several parameters remain fixed across all conditions: the underlying corpus (a localized subset of 500 questions extracted from the StackOverflow full dataset), the core dialogue LLM (Qwen3.6-35B-A3B), the embedding encoder (Qwen3-Embedding 8B), the initial cluster count derived via K-means (used to enrich the system prompt of the interview agent), and the base behavioral persona of the simulated user.

3.2 Outcomes and Hypothesis Testing

We categorize our experimental objectives into primary and secondary classifications to maximize statistical transparency.

- **Primary Outcome:** *Turns to convergence* serves as our primary outcome metric. It provides an objective, direct evaluation of dialogue efficiency and establishes the headline criteria for determining which question paradigm minimizes human friction.
- **Secondary Outcomes:** Supporting metrics include the total tokens exchanged, user response tokens, and the mean faithfulness score (the composite average of coherence, alignment, and separation metrics). These support or clarify structural anomalies observed in the primary outcome but do not dictate the headline claim.

We formally evaluate the following hypothesis:

Multiple-choice questions represent the optimal operational tradeoff between user cognitive load (response complexity) and information gained per turn, leading to superior efficiency compared to yes/no or open-ended configurations.

3.3 Experimental Matrix

To test this hypothesis, we execute $N = 108$ distinct experimental runs. The three question types are balanced uniformly, resulting in 36 independent trials per interactive condition. The simulation environment coordinates a matrix of 18 unique variations, blending 3 distinct user personas with 6 target goals. Each specific combination is executed exactly 2 times to ensure statistical reliability while mitigating runtime noise.

3.4 Decision Rule

To establish statistical significance without relying on rigid parametric assumptions, we employ a non-parametric bootstrapping strategy. For each pairwise comparison between interactive question conditions (e.g., Multiple-Choice vs. Open-Ended), we calculate the absolute difference in sample means and generate a 95% bootstrap confidence interval using 10000 resamples. We conclude that Condition A is significantly more efficient than Condition B if and only if the 95% confidence interval for the mean difference ($A - B$) strictly excludes zero.

3.5 Baseline and Experimental Limitations

The baseline condition embeds documents using the Qwen3-Embedding model under standard uninstructed configurations, skipping the interview stage and omitting task-specific metadata from the encoder prompt.

This introduces an architectural confound noted explicitly in our protocol: the baseline differs from experimental conditions by both the *absence of user preference elicitation* and the *absence of task instructions during embedding generation*. As a result, our study design cannot mathematically isolate whether an improvement in downstream cluster faithfulness is uniquely driven by dialogue elicitation, instruction-guided embedding mechanics, or their combined synergy. Isolating these independent effects remains an objective for future iterations.

3.6 Simplifying Operational Assumptions

To maintain a mathematically tractable and structured evaluation framework, the experimental design operates under four foundational operational assumptions:

- **User Responsiveness:** Simulated users are assumed to be fully responsive. When prompted, the agent always provides an explicit preference or definitively assigns a datapoint to a cluster. Conversational abstentions, ambiguous inputs, or “I do not know” responses are explicitly out of scope.
- **Preference Consistency:** User preferences are modeled as perfectly stable within a single conversational session. The user agent will not contradict its earlier responses or exhibit shifting criteria mid-loop.
- **Modality Restriction:** The system architecture operates strictly on a text-only data matrix. Multimodal document processing and heterogeneous data inputs are considered out of scope.
- **Offline Clustering Topology:** The framework calculates a single, absolute clustering configuration at the conclusion of the dialogue loop. Partitioning is not executed incrementally during active conversation.

4 Methodology

The architecture of *Cluster Around and Find Out* operates as an interactive pipeline that translates qualitative natural language conversations into quantitative geometric adjustments within a vector space. The workflow is divided into four main phases: initial corpus overview and baseline specification, prompt-engineered preference elicitation, instruction-guided embedding generation, and final customized clustering and evaluation pipeline. The system couples conversational efficiency tracking with multi-dimensional metric verification to evaluate end-to-end user alignment.

4.1 Phase 1: Initial Corpus Overview and Baseline Specification

The pipeline begins with an unlabelled text corpus consisting of a subset of 500 questions from the StackOverflow full dataset. Before initializing the user interface, the system generates an initial structural layout of the text space to gather clean context for the conversational agent.

To achieve this, the corpus is processed using a standard, conventional text embedding system. A traditional K -means clustering algorithm is executed over these conventional vectors with K arbitrarily set to 10. From each of the 10 identified clusters, the system randomly samples 3 representative documents. This yields a total of 30 anchor documents that encapsulate the broad thematic variation of the dataset. These 30 documents are fed into the system prompt of the conversational agent to ground its domain awareness.

Crucially, this phase also defines the unstructured baseline used for downstream evaluation. For the baseline condition, the entire 500-document dataset is embedded using the primary encoder, **Qwen3-Embedding (8B)**, configured with a static instruction stating that the user has no specific sorting preferences. The downstream clustering algorithm is then selected and configured by **Qwen3.6-35B-A3B** without any human alignment conditioning. This baseline assignment is completely isolated from the conversation pipeline; it is never shown to the user or used by the interview agent, serving strictly as a control group to evaluate preference alignment post-experiment.

4.2 Phase 2: Prompt-Engineered Preference Elicitation

Instead of relying on heavy optimization or post-training adjustments, our conversational interface relies entirely on strategic prompt engineering. The system initializes the interview agent using **Qwen3.6-35B-A3B**. The model is configured via a comprehensive system prompt that embeds:

1. The operational role of an expert data-mining interviewer.
2. The dataset context derived from the Phase 1 sampling, passing the 30 randomly sampled anchor documents directly into the prompt.
3. Strict formatting rules corresponding to one of the three tested interactive paradigms: *yes/no*, *multiple-choice*, or *open-ended* question formats.

During the dialogue loop, the agent interviews a simulated user configured with specific combinations of 3 personas and 6 goals. The agent presents questions regarding document relationships, target granularity, and sorting logic, utilizing the initial cluster summaries to present concrete examples. The interaction continues dynamically until the conversation hits convergence criteria—either reaching a structural consensus or hitting an arbitrary max turn limit set to 20. At this point, the Qwen3.6-35B-A3B agent synthesizes the entire chat history into a concise, dense paragraph of text: the *User Preference Profile*.

4.3 Phase 3: Instruction-Guided Embedding Generation

Once the User Preference Profile is synthesized, it is directly injected into the downstream embedding pipeline as a formatting instruction. We leverage **Qwen3-Embedding (8B)**, a decoder-only model optimized for instruction-tuned vector pooling.

Unlike traditional static encoders that project identical text to identical vectors regardless of intent, the Qwen3-Embedding model accepts a structured instruction prefix. We construct our embedding prompt by mapping the User Preference Profile directly into the instruction field of the encoder. The raw StackOverflow questions are then passed to the model under this prefix. The model utilizes its internal bidirectional attention mechanisms to alter the weights of semantic tokens dynamically based on the

instruction. This maps the text into an adjusted high-dimensional space where documents sharing user-preferred traits are projected closer together, and documents violating the user’s criteria are pushed further apart.

4.4 Phase 4: Final Customized Clustering and Evaluation Pipeline

In the final phase, the system applies a clustering algorithm over the newly adjusted instruction-guided embedding space. Because the vector geometry itself has been altered by the Qwen3-Embedding model to reflect the conversation’s nuances, a standard geometric partitioning algorithm can now capture subjective human intents.

The resulting clusters represent the customized configuration tailored to the user’s specific persona and goal. To maintain strict experimental alignment, the performance of these final clusters, along with the operational metrics of the dialogue loop, are captured, recorded, and processed using our LLM-as-judge. This allows for an end-to-end evaluation of how well the interactive prompting paradigm shifts the underlying mathematical structure of the data compared to our uninstructed Phase 1 control baseline.

5 Results

This section details the empirical findings from our $N = 108$ balanced simulation runs, computed via 10,000 bootstrap resamples at a 95% confidence interval (seed = 2083521328). We evaluate the operational dynamics of our interactive paradigms against their respective uninstructed baselines (Stage 1) and conduct an absolute comparison between the question types (Stage 2).

5.1 Stage 1: Interactive Conditions vs. Uninstructed Baselines

The baseline conditions bypass conversational alignment entirely, meaning they register zero values for turns, dialogue tokens, and user response text. The primary value of the baseline

lies in establishing a control threshold for downstream cluster geometry.

As summarized in Table 1, every interactive question type required a statistically significant volume of conversational exchange compared to the instantaneous control baseline. However, evaluating the secondary outcome reveals an unexpected geometric convergence. Across all three tested formats—yes/no, multiple-choice, and open-ended—the resulting mean faithfulness scores do not manifest a statistically significant delta over their respective uninstructed control baselines.

The bootstrapped 95% confidence intervals for the mean difference (Condition – Baseline) all encompass zero:

- **Yes/No Faithfulness Delta:** +0.222, 95% CI: $[-0.148, +0.602]$ (Not Significant).
- **Multiple-Choice Faithfulness Delta:** +0.037, 95% CI: $[-0.333, +0.389]$ (Not Significant).
- **Open-Ended Faithfulness Delta:** +0.231, 95% CI: $[-0.120, +0.593]$ (Not Significant).

This indicates that within the parameters of this pipeline, the geometric separation and alignment modifications generated by the instruction-tuned embedding layer did not yield a mathematically distinct advantage over blind baseline assignments.

To provide a more granular diagnostic look into these overall performance patterns, Table 2 isolates and breaks down the three underlying dimensions—coherence, alignment, and separation—that compile the aggregate mean faithfulness ratings.

5.2 Stage 2: Pairwise Question Type Comparison

To determine the optimal human-computer interface, we analyze the absolute variations between the interactive treatments across our observed criteria.

5.2.1 Turns to Convergence (Primary Outcome)

The primary efficiency metric reveals a highly distinct separation between paradigms, isolating an absolute hierarchical rank. Multiple-choice questions proved to be the most efficient strategy, converging in a mean of just 7.83 turns. Open-ended queries followed with a mean of 9.64 turns, while yes/no interactions required the most extensive dialogues, averaging 12.97 turns to convergence.

Pairwise bootstrapping confirms the statistical significance of this ranking:

- **Yes/No vs. Multiple-Choice:** Multiple-choice dramatically outperforms yes/no (Diff = +5.139, 95% CI: $[+3.111, +7.250]$).
- **Yes/No vs. Open-Ended:** Open-ended queries converge significantly faster than binary choices (Diff = +3.333, 95% CI: $[+1.028, +5.583]$).
- **Multiple-Choice vs. Open-Ended:** Multiple-choice establishes a statistically verified lead over open-ended configurations (Diff = -1.806 , 95% CI: $[-3.556, -0.139]$).

5.2.2 Token Consumption and Question Clarity

The volume of language exchanged tracks the structural density of the interaction profiles. Total tokens exchanged exhibited no statistically significant variation between the yes/no framework (1238.81) and the multiple-choice framework (1196.42), with a bootstrapped difference interval of $[-241.974, +337.619]$, which strictly encompasses zero. Similarly, open-ended token generation (1383.81) fell within a non-significant baseline margin compared to its peers.

However, the user response token count—serving as our direct proxy for cognitive load and question clarity—reveals massive, statistically significant variations. Multiple-choice questions required a sparse mean of 100.92 tokens from the user. Yes/no questions demanded significantly more descriptive clarifications from the simulated agent,

Condition Framework	Turns to Convergence	Total Tokens	User Tokens	Mean Score
Yes/No Condition	12.97	1238.81	338.69	2.20
<i>Yes/No Baseline</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>1.98</i>
Multiple-Choice	7.83	1196.42	100.92	2.23
<i>Multiple-Choice Base</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>2.19</i>
Open-Ended Condition	9.64	1383.81	616.50	2.19
<i>Open-Ended Baseline</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>1.96</i>

Table 1: Aggregated Dialogue Dynamics and Mean Faithfulness Scores Across Interactive Frameworks ($N = 108$ Runs)

Condition Framework	Coherence Score	Alignment Score	Separation Score
Yes/No Condition	2.58	1.75	2.28
<i>Yes/No Baseline</i>	<i>2.44</i>	<i>1.58</i>	<i>1.92</i>
Multiple-Choice	2.67	1.72	2.31
<i>Multiple-Choice Base</i>	<i>2.47</i>	1.78	2.33
Open-Ended Condition	2.61	1.75	2.22
<i>Open-Ended Baseline</i>	<i>2.44</i>	<i>1.42</i>	<i>2.03</i>

Table 2: Granular Breakdown of LLM-as-Judge Faithfulness Sub-Metrics

averaging 338.69 tokens. Open-ended prompts generated the heaviest linguistic overhead, forcing the user to draft an extensive mean of 616.50 tokens per run.

Pairwise testing confirms that multiple-choice queries minimize user token generation over yes/no (Diff = +237.778, 95% CI: [+150.722, +332.640]), while yes/no interfaces restrict response lengths significantly more effectively than unstructured open questions (Diff = -277.806, 95% CI: [-406.444, -151.358]).

5.2.3 Downstream Faithfulness Alignment

In contrast to the highly dynamic efficiency metrics, the resulting mean faithfulness scores remain entirely uniform across the interactive strategies. The composite mean scores reached 2.20 for yes/no, 2.23 for multiple-choice, and

2.19 for open questions. The corresponding sub-metrics—coherence, alignment, and separation—maintain flat distributions across all parameters. Pairwise bootstrapping verifies that changing the question style yields absolutely no statistical difference in final cluster faithfulness.

5.3 Evaluation Framework Validation: Inter-Rater Agreement

To assess the reliability and operational validity of our downstream LLM-as-a-judge framework, a human-in-the-loop cross-validation check was executed. A human expert independently annotated a random sample of clustering configurations ($N = 6$ items, yielding 18 total individual metric ratings) across the three core evaluation dimensions. Inter-rater agreement between the human expert and the automated LLM judge was quantified using a quadratically

weighted Cohen’s Kappa (κ). To mathematically account for the restricted sample volume, a 10,000-resample bootstrap was applied to calculate 95% confidence intervals. The agreement metrics are detailed in Table 3.

The primary composite Mean Score yielded a $\kappa = 0.571$, indicating a *moderate* level of baseline agreement between human judgment and the automated evaluator. Evaluated individually, the system exhibited *substantial* agreement on foundational text traits such as semantic coherence ($\kappa = 0.690$) and profile alignment ($\kappa = 0.640$). Conversely, spatial separation achieved only *fair* agreement ($\kappa = 0.385$), capturing the inherent subjectivity involved in contextually isolating overlapping technical terms.

Due to the high operational cost of expert manual annotations, this validation was strictly restricted to an exploratory sample ($N = 6$). Consequently, the resulting bootstrap confidence intervals are expectedly wide, and these coefficients must be interpreted strictly as an indicative sanity check of judge behavior rather than an absolute population statistic. Nevertheless, the moderate agreement observed across the primary score configurations confirms that the automated evaluator operates as a functional proxy for tracking relative performance trends across our primary simulation matrix.

6 Discussion

The empirical findings gathered from our $N = 108$ balanced simulation matrix offer clear insights into interactive text exploration pipelines, unequivocally supporting our primary research hypothesis.

6.1 Validation of the Primary Hypothesis

Our core hypothesis stated that multiple-choice questions represent the optimal operational tradeoff between user effort and information gained per turn. The experimental results completely substantiate this claim. Multiple-choice frameworks achieved the absolute premier rank

in conversational efficiency, converging in only 7.83 turns.

Crucially, our clarity metrics unveil the mechanical reasons behind this success. While yes/no selections seem simple, they lack informational bandwidth, forcing the system into protracted binary searching that prolonged the dialogue to a mean of 12.97 turns. Conversely, while open questions maximize the information harvested per turn, they trigger highly verbose replies (616.50 tokens on average), indicating a high cognitive burden.

Multiple-choice questions resolve this tension. They compress user output to a minimal 100.92 tokens while providing rich semantic boundaries. This allows the interviewing agent to isolate user intent with minimal conversational turns, striking an optimal balance between low user effort and rapid informational gain.

6.2 The Faithfulness Paradox

The most surprising revelation of our study is the uniform lack of significant difference in downstream cluster faithfulness, both across the question types and against the uninstructed baseline control. This uniform plateau (hovering between 1.96 and 2.23 out of 5.00) points to a critical systemic limitation.

While the Qwen3.6-35B-A3B agent successfully extracted diverse preferences, translating these qualitative texts into instruction prompts did not yield localized vector space configurations that an LLM judge recognized as highly faithful. This implies that instruction-guided text embeddings may have upper resolution boundaries when adjusting vector distances for highly niche, domain-specific corpora like Stack-Overflow questions. The instruction space might handle general classification adjustments well, but struggles to cleanly separate overlapping technical terms based purely on brief prompt prefixes.

Moreover, extremely concise text strings—such as Stack Overflow question titles—may inherently lack the semantic density required to effectively condition the downstream encoder, thereby limiting its capacity to

Evaluation Metric	Items Rated	Weighted Cohen’s κ	95% Bootstrap CI
Mean Score ^a	6	0.571	[−0.667, +0.909]
Coherence	6	0.690	[+0.000, +0.889]
Alignment	6	0.640	[+0.000, +0.909]
Separation	6	0.385	[−0.500, +0.769]

^a Calculated as the composite mean: (coherence + alignment + separation)/3

Table 3: Inter-Rater Agreement Metrics (Weighted Cohen’s κ) between Human Expert and LLM Judge

generate highly aligned embedding vectors.

6.3 Synergy with Existing Literature

Our results build directly upon foundational concepts within preference learning. By demonstrating that a prompt-engineered Qwen3.6-35B-A3B model can efficiently drive conversational convergence without specialized model tuning, we validate a lean alternative to the heavy trajectory optimizations found in frameworks like TO-GATE [4].

However, the fact that our final cluster modifications flattened relative to the baseline provides a stark contrast to systems like Cluster-LLM [1] or Dial-In LLM [2]. Those frameworks actively enforce hard pairwise geometric constraints or utilize an active LLM looping mechanism to force partition rearrangements. Our pipeline highlights that relying solely on a downstream instruction-tuned encoder—such as Qwen3-Embedding—presents an engineering bottleneck: it trades away aggressive, computationally intensive cluster refinement for speed and scalability, but sacrifices nuanced, human-aligned faithfulness in the process.

7 Limitations

While the empirical findings from our $N = 108$ balanced simulation matrix offer clear architectural insights, several distinct technical and structural boundaries qualify the generalizability of this study.

7.1 Data Materiality and Information Bottlenecks

A primary constraint stems from the inherent structural attributes of the evaluation corpus under our text-only modality restriction. The text fragments extracted from the StackOverflow dataset consist almost exclusively of extremely concise question titles. Such brief text strings introduce a severe informational bottleneck. Truncated inputs lack the semantic density, syntax variation, and contextual richness required to deeply steer the internal bidirectional attention mechanisms of a decoder-only embedding model. Consequently, even when paired with an explicitly synthesized instruction prefix, the encoder is given very little baseline context to reshape, limiting its ability to project structurally distinct vector coordinates.

7.2 Granularity and Specialized Domain Resolution

The evaluation pipeline relies on a highly specialized technical dataset containing subtle conceptual overlaps (e.g., adjacent programming framework questions). The uniform plateau observed in our downstream faithfulness scores across both active conditions and uninstructed baselines indicates a systemic limitation in the resolution boundary of instruction-guided vector tuning. Static, brief instruction prefixes—like our generated User Preference Profiles—appear capable of driving coarse-grained global adjustments (such as broad thematic sorting), but fail to supply the heavy mathematical steering required to cleanly separate overlapping tech-

nical terms or enforce highly nuanced, human-specified partitioning criteria.

7.3 Algorithmic Staticity and Lack of Direct Refinement

Our methodology explicitly isolates the conversation pipeline from the embedding generation phase to test a lean, lightweight alternative to model fine-tuning, resulting in an offline clustering topology. However, this architectural choice creates a unidirectional information flow: the qualitative conversation ends, a profile is generated, and the encoder runs exactly once. The system lacks a dynamic feedback loop or active geometric constraint enforcement, such as forcing direct pairwise document distances or iteratively re-clustering based on immediate human corrections. This design ensures computational efficiency and speed, but sacrifices the iterative, hyper-targeted alignment corrections seen in more resource-heavy preference-learning frameworks.

7.4 Simulated User Persona Modeling

The dialogue loop was evaluated using simulated user agents operating under strict persona and goal matrices. While this programmatic simulation allowed us to execute a rigorously balanced and fully reproducible matrix of 108 identical condition runs, it introduces an inevitable gap between simulated behavior and authentic human exploratory patterns. Real-world human users frequently exhibit conversational drifts, inconsistent sorting terminology, and shifting goals mid-dialogue, which would likely test the convergence mechanics and token efficiency of the conversational agent in ways not fully captured by automated personas.

7.5 Impact of Behavioral and Structural Assumptions

The empirical boundaries of our evaluation are further qualified by the simplifying operational assumptions declared at the conclusion of our study design (see Section 3.6). Specifically, the

assumptions of *user responsiveness* and *preference consistency* eliminate the conversational volatility typical of deployment with human subjects. Real-world users frequently express ambiguity, offer contradictory feedback, or choose to abstain from forcing a preference when presented with unfamiliar technical clusters. By isolating the system from noisy, inconsistent, or non-responsive dialogue inputs, our current baseline benchmarks likely reflect an idealized efficiency curve. Consequently, further evaluation is required to determine how robustly the conversational agent handles edge cases where human intent drifts or fails to cooperate with prompt constraints.

7.6 Limited Human-in-the-Loop Validation

Finally, while the LLM-as-a-judge metric architecture was successfully cross-validated against human expert ratings, the manual annotation pool was heavily constrained by project timelines and resource availability. Although the moderate composite agreement ($\kappa = 0.571$) validates the judge as a functionally sound baseline proxy for our comparative matrix ($N = 108$), a larger and more diverse human-annotated sample would be required to fully calibrate the evaluator. Without extensive validation across an expanded corpus, the current kappa estimates remain statistically unstable, masking potential edge cases where automated multidimensional scoring might diverge from highly idiosyncratic human perceptions of cluster quality.

8 Ethics and Broader Impact

This section addresses the ethical considerations, dataset provenance, and algorithmic transparency measures associated with this research framework.

8.1 Dataset Provenance and Content Integrity

Our experimental evaluation relies entirely on an open-source, publicly accessible dataset com-

piled from scraped question titles on the Stack Overflow platform. Because the corpus is strictly restricted to high-level technical inquiries and programming titles rather than user account profiles, it contains no personally identifiable information. Consequently, no secondary data anonymization pipelines were required. Furthermore, a rigorous manual review identified no systemic demographic or sociocultural biases within this technical text domain that could skew or harm the validity of our downstream clustering evaluations.

8.2 Declaration of Generative AI Assistance

In accordance with contemporary academic integrity standards and institutional disclosure guidelines, we explicitly declare the strategic utilization of generative artificial intelligence (AI) tools throughout this project’s lifecycle.

Large Language Models (LLMs) were leveraged under strict human supervision across three distinct domains:

1. **Conceptual Brainstorming and Troubleshooting:** Generative tools were utilized as an analytical sounding board to brainstorm architectural alternatives and troubleshoot complex structural bottlenecks that emerged during the pipeline’s development phases.
2. **Software Development Support:** Automated code-generation tools were utilized to accelerate script compilation, optimize syntax for our bootstrapping algorithms, and refine data-parsing pipelines.
3. **Manuscript Synthesis and Refinement:** Generative text assistants were employed during the drafting phase to enhance grammatical precision, eliminate typographic errors, and assist in restructuring complex prose into a cohesive, formal academic tone.

Crucially, all primary conceptual designs, final algorithmic architectures, simulation parameter boundaries, and empirical interpretations

were conceived exclusively by the human authors. The generative models functioned purely as structural, technical, and developmental optimization assistants, and all final outputs were rigorously audited, verified, and edited for factual validity prior to inclusion.

9 Conclusion and Future Work

This study introduced and systematically evaluated an interactive, dialogue-driven preference extraction framework designed to bridge the gap between qualitative human clustering intent and automated vector space partitioning. By deploying a Qwen3.6-35B-A3B user-facing agent across a rigorously balanced simulation matrix, we demonstrated that conversational interaction can successfully surface distinct thematic sorting goals without the resource-heavy overhead of direct model fine-tuning.

Our empirical evaluation uncovered key trade-offs between conversational elicitation frameworks and downstream embedding adjustments. While multiple-choice structures proved significantly more efficient—minimizing conversational turns to convergence (7.83 turns) and reducing total token consumption—open-ended dialogues offered unconstrained structural expression at the cost of higher linguistic volatility. However, across all active interactive conditions, the resulting instruction-guided text embeddings encountered a distinct resolution boundary. When applied to hyper-short, domain-specific text fragments like Stack Overflow question titles, static prompt prefixes yielded minor global shifts but struggled to deeply restructure local vector distances or cleanly separate overlapping technical terms.

These findings suggest that while conversational preference extraction is highly viable for capturing macro-level human intent, lightweight instruction-based embedding modulation faces an informational bottleneck when processing short text snippets. Relying purely on a single-shot, offline clustering topology limits the system’s capacity to enforce fine-grained topological constraints.

9.1 Future Work

Building upon the insights and structural boundaries identified in this work, we propose three primary directions for future research:

1. **Dynamic Geometric Feedback Loops:** Future iterations should transition from an offline topology to an online, closed-loop interactive architecture. By feeding downstream clustering configurations back into the dialogue loop, the agent could dynamically enforce pairwise constraints in real time based on immediate user feedback.
2. **Adaptive Context Expansion:** To resolve the informational bottleneck observed with short text strings, future frameworks should explore automated context augmentation. Automatically parsing and appending supplementary metadata—such as generated keywords or keyphrases—could provide the semantic density required for embedding models to execute finer-grained separations.
3. **Evaluation under Dynamic Cognitive Loads:** Finally, deploying this framework with real-world human user studies will be essential to evaluate system robustness against conversational drift, ambiguous criteria, and non-responsive edge cases, moving beyond the idealized behavior enforced by our simulated user personas.

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